

Functional Knee Brace Alters Predicted Knee Muscle and Joint Forces in People With ACL Reconstruction During Walking

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Functional knee braces used during rehabilitation from injury and surgery to the anterior cruciate ligament (ACL) have been reported to provide a strain-shielding effect on the ACL in healthy people while standing, reduce quadriceps electromyography in ACL-deficient individuals, and alter joint torque patterns in people with ACL reconstruction during walking. These results led to the hypothesis that functional knee braces protect a reconstructed ACL during dynamic activity by reducing the anterior shear load applied to the knee. This hypothesis was tested by investigating the effects of a functional knee brace on lower extremity muscle forces and the anteroposterior shear force at the knee joint during the stance phase of walking in people with ACL reconstruction. Ground reactions and sagittal plane video were recorded from 9 ACL-reconstructed individuals as they walked with and without a functional knee brace, and from 10 healthy people without the functional knee brace. Inverse dynamics were used to calculate the net joint torques in the lower extremity during the stance phase. Hamstrings, quadriceps, and gastrocnemius muscle and knee anteroposterior shear force were then predicted with a sagittal-plane mathematical model. Compared to healthy individuals, those with ACL reconstruction walked with 78% more hamstrings impulse and 19% less quadriceps impulse (both $p < .05$). The functional knee brace produced an additional 43% increase in hamstrings impulse and an additional 13% decrease in quadriceps impulse in the ACL group. Peak anterior knee shear force and anterior impulse were 41% lower and 16% lower in ACL vs. healthy individuals, respectively. The functional knee brace further reduced the peak knee shear force and impulse 28% and 19%, respectively, in the ACL group. It was concluded that a functional knee brace protects a reconstructed ACL during walking by altering muscle forces and reducing the anterior shear force applied to the knee joint.

Key Words: gait, locomotion, injury, biomechanical model

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Introduction

The principle function of the anterior cruciate ligament (ACL) is to constrain anterior motion of the tibia relative to the femur within physiological limits. While the ACL is relatively strong and can withstand forces reaching approximately 2,200 N in laboratory tests (Woo, Hollis, Adams, Lyon, & Takai, 1991), it is often injured in a variety of human activities. Subsequent reconstruction surgery for ruptured ACL most often uses the central one-third of the patellar tendon (i.e., bone-patellar tendon-bone) as the replacement tissue for the ligament. While this surgery produces a structurally sound knee joint, the injury and surgery also necessitate functional neuromuscular adaptations (DeVita, Hortobagyi, & Barrier, 1998a; Kowalk, Duncan, McCue, & Vaughan, 1997), reduce joint proprioception (Co, Skinner, & Cannon, 1993), and compromise long-term health (Jarvela, Paakkala, Kannus, & Jarvinen, 2001).

Functional knee braces are often used during rehabilitation from ACL injury and surgery in order to protect the healing graft and knee joint. This protective function may be achieved through the mechanical constraint caused by the functional knee brace on the knee joint, and by alterations in neuromuscular function while wearing a brace. Beynnon, Johnson, Fleming, et al. (1997) measured ACL strain in response to anterior shear forces applied to the tibia in healthy individuals who stood with their knees flexed 30°. With forces reaching 140 N, the ACL strain was reduced approximately 50% while wearing a functional knee brace vs. not wearing the brace. Beynnon et al. concluded that the functional knee brace provided a strain-shielding effect on the ACL.

Neuromuscular adaptations to functional knee braces have been shown through changes in muscle electromyography (EMG) and joint torques during dynamic activities. Bracing reduced quadriceps and hamstrings reaction times by approximately 20%, as measured with EMG during isokinetic testing (Wojtys, Kothari, & Huston, 1996), and reduced peak muscle activity during a side-cutting movement (Branch, Hunter, & Donath, 1989) in ACL-deficient individuals. For people with ACL reconstruction, the functional knee braces reduced the net extensor torque at the knee and increased the net extensor torque at the hip during walking 3 weeks after surgery (DeVita, Lassiter, Hortobagyi, & Torry, 1998b).

The observed strain-shielding effect in standing, healthy people, the reduced quadriceps EMG in ACL-deficient individuals, and the altered joint torque patterns in those with ACL reconstruction led us to hypothesize that functional knee braces protect a reconstructed ACL during dynamic activity by reducing the load applied to the knee. To test this hypothesis, we investigated the effects of a functional knee brace on lower extremity muscle forces and the anteroposterior shear force at the knee joint during the stance phase of walking in individuals who had recently undergone ACL reconstruction and in healthy people. The research protocol was designed to answer the following questions pertaining to the stance phase of walking:

- Do muscle forces differ between healthy people and those who have had ACL reconstruction?
- Does a functional knee brace change muscle forces in people with ACL reconstruction?
- Does a functional knee brace change the anteroposterior knee joint force in people with ACL reconstruction?

Muscle and joint forces were predicted with a mathematical model developed specifically for human locomotion.

Methods

Nine young adults who had undergone ACL reconstruction (mean age 22.1 yrs, *SD* 4.7; mass 73.6 kg, *SD* 6.9) volunteered for the study. None had ever had a previous knee injury. Each one had arthroscopically assisted, bone-patellar tendon-bone reconstruction surgery approximately 3 weeks after injury. Hyperextension and full weight-bearing were achieved by 7 to 10 days post op. Closed kinetic chain exercises were started during the 2nd to 3rd week to strengthen the hamstrings and quadriceps. The participants began jogging between the 4th and 5th weeks; the rehabilitation program lasted 6 months. Serving as a control group were 10 healthy young adults (mean age 22.7 yrs, *SD* 2.6; mass 75.2 kg, *SD* 10.3) with no history of lower limb injury, who were tested without a brace. All gave informed consent prior to participation.

Ground reaction forces during the stance phase of walking were measured at 1,000 Hz with a force platform (AMTI, model LG6-4-2000) located in the center of a 20-m level walkway. Sagittal plane kinematics of the stance phase were recorded with a video camera (SONY Corp., Model SSC M350) and video cassette recorder (JVC, model HRS5100U) at 60 Hz. An infrared system (Lafayette, model 63520) was used to measure the time it took each participant to walk a 6-m distance over the platform.

Those with ACL reconstruction were tested in an off-the-shelf functional knee brace (C.Ti. Edge functional knee brace, Innovation Sports, Irvine, CA) which was applied by the same researcher to each of them. The brace weighed 6.9 N and had a multicentric hinge, rigid thigh and leg shell components, and velcro strapping to secure it to the limb. The adjustable hinge was set at a 10° extension stop. The control group was not tested in the brace.

All participants wore athletic shoes, a T-shirt, and bicycle shorts during the test. Those with ACL reconstruction were tested approximately 39 days after surgery (range = 35 to 42 days). All were weighed, and circumferences were obtained of the upper thigh, knee, ankle, and metatarsal heads on the injured limb in the ACL group, or on the right limb in the control group. Reflective markers were then placed on the lateral side of the shoe at the head of the fifth metatarsal and heel, and on surface locations over the lateral malleolus, lateral femoral condyle, greater trochanter and shoulder on the injured side, or right side in the control group. A reflective marker was also placed on the hinge of the functional knee brace so that it was directly over the marker on the femoral condyle. The ACL group practiced walking with and without the brace to become acclimated to the conditions and walkway prior to testing, while the control group practiced without the brace. Participants walked at their own chosen speed. Eight trials of data were collected, on average, for each individual and condition. Both groups walked at similar speeds, 1.45 m/s for the ACL group vs. 1.51 m/s for the healthy controls.

Sagittal plane Cartesian coordinates of the reflective markers were derived throughout the stance phase using the Peak Performance system (Peak Performance Technologies, Inc., Englewood, CO). High frequency error was removed from the digitized coordinates with a low-pass digital filter that automatically selected the cut-

off frequency (Winter, 1990). The mean cutoff frequency was about 5 Hz. Linear position data were interpolated to 200 Hz using a cubic spline routine without further smoothing so that the kinetic and kinematic data could be synchronized. Angular positions were calculated at the hip, knee, and ankle joints.

Inverse dynamics with linear and angular Newtonian equations of motion were used to calculate the reaction forces and torques at each joint throughout the stance phase. The lower extremity was modeled as a rigid, linked segment system. Magnitude of the segmental masses, their moments of inertia, and the locations of the mass centers were estimated from the position data using a mathematical model (Hanavan, 1964), segmental masses reported by Dempster (1959), and the individual's own anthropometric data. Center of pressure in the anteroposterior direction was calculated from the ground reaction forces and the mediolateral moment on the force platform. The accuracy of the center of pressure was tested by applying a force of 550 N at seven known locations on the platform. The average error in the anteroposterior direction was 0.003 m. Error of this magnitude will produce approximately 4% error in the prediction of lower extremity joint torques (McCaw & DeVita, 1995). The inverse dynamic analysis included applying the ground reaction forces to the foot at the center of pressure, calculating the vertical and horizontal joint reaction forces at the ankle, and calculating the joint torque at the ankle. The joint torque was expressed as a reaction torque to all external torques and therefore represented the internal torque produced by the muscles and other tissues crossing the joint. The ankle forces and torque were calculated with these equations:

$$F_z = ma_z + mg - GRF_z \quad (1)$$

$$F_y = ma_y - GRF_y \quad (2)$$

$$T = I\alpha - GRF_z(d1) - GRF_y(d2) - F_z(d3) - F_y(d4) \quad (3)$$

where z and y are vertical and horizontal directions, F and GRF are the ankle joint and ground reaction forces, m and I are the mass and moment of inertia of the foot, a and α are the linear and angular accelerations of the foot, T is the ankle joint torque, $d1$ to $d4$ are the horizontal and vertical lever arms between the ankle joint or ground reaction forces and the center of mass of the foot, and g is the acceleration due to gravity. The reaction forces and torque at the ankle were then reversed and applied to the leg and the process was repeated for the knee forces and torque. The knee kinetics were then used to calculate the hip forces and torque.

Forces in the hamstrings, quadriceps, and gastrocnemius muscles and the anteroposterior shear force at the knee were calculated with a mathematical model (Figure 1). The tibia was defined as the line between the ankle and knee positions and the femur was defined as the line between knee and hip positions. Anterior shear force at the knee was a force directed perpendicular and anterior to the tibia. Force in the triceps surae was determined from the plantarflexor torque at the ankle joint, assuming no co-contraction by the dorsiflexors. This method was used previously (Scott & Winter, 1990; Simonsen, Dyhre-Poulsen, Voigt, et al., 1995) and was also supported by the literature which showed a strong association between plantar flexor torque and gastrocnemius and soleus EMG (Anderson & Pandy, 2001; Hof, Pronk, & van Best, 1987) and low anterior tibialis EMG during the stance phase (Simonsen et al., 1995).

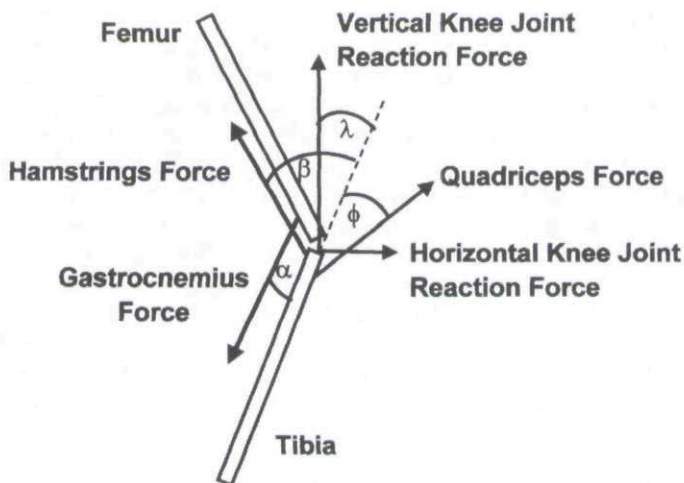


Figure 1 — Schematic of mathematical model of muscle and knee joint forces. Gastrocnemius and hamstrings muscle forces are calculated, then applied to the knee at angles α and β to the tibia to calculate their flexor torque. Quadriceps force is then calculated from the net torque and this flexor torque. Quadriceps force is applied at angle ϕ to the tibia. All forces and the joint reactions are applied to the tibia to calculate anteroposterior shear force at the knee.

Gastrocnemius force (G) was calculated as a function of its physiological cross-sectional area (PCA) relative to the total PCA of the triceps surae muscles (i.e., 0.319; Yamaguchi, Sawa, Moran, et al., 1990):

$$G = At / ATd * 0.319 \quad (4)$$

where At was the ankle torque and ATd was the lever arm for the Achilles tendon. Lever arm values for each calculation throughout the stance phase were based on the observed angular position of the ankle joint and were derived by digitizing lever arm-ankle joint position curves from the literature (Fukunaga, Roy, Shellock, Hodgson, & Edgerton, 1996; Klein, Mattys, & Rooze, 1996; Rugg, Gregor, Mandelbaum, & Chiu, 1990; Spoor, van Leeuwen, Meskers, Titulaer, & Huson, 1990), interpolating each curve to 1° increments with a cubic spline, and averaging the curves through the range of 20° dorsiflexion to 30° plantar flexion. The mean value for ATd throughout the stance phase was 0.051 m. This technique was used for all muscle lever arms.

The direction of G was determined from the heel and knee marker positions and was expressed as α , the angle between G and the tibia. The heel marker was used to represent the distal end of the gastrocnemius. The proximal end was positioned 0.020 m superior and 0.023 m posterior to the knee joint, along the line of the femur (Seireg & Arvikar, 1973). The resultant direction of G was, on average, about 3° from parallel with the tibia, thus this force applied a small posterior shear load at the knee.

Hamstrings force was calculated from the extensor torque at the hip joint observed generally during the first half of stance. This method was supported by the literature that showed a strong association between hip extensor torque and hamstrings EMG in early stance (DeVita, 1994; Glitsch & Baumann, 1997). The hip extensor torque was assumed to be produced by the hamstrings and gluteus maximus muscles without co-contraction of the hip flexors during the first half of stance. This assumption was generally supported by EMG measures and muscle force predictions in the literature, except that rectus femoris does contract and produce force during some of this period (Anderson & Pandy, 2001; Glitsch & Baumann, 1997). Predicted estimates of rectus femoris force during the first half of stance are low, however (Glitsch & Baumann, 1997), producing a low amount of error for hamstrings force. The predicted hamstrings force accounted for both the hamstrings PCA relative to the total PCA of the hamstrings and gluteus maximus and the hamstrings lever arm at the hip relative to the gluteus maximus lever arm. The total hamstrings proportion, H_p , to the hip torque was calculated as:

$$H_p = [\text{Ham PCA} / (\text{Ham PCA} + \text{GM PCA})] (H_d / \text{GMd}) \quad (5)$$

where Ham PCA and GM PCA were the hamstrings and gluteus maximus PCAs, and H_d and GMd were the hamstrings and gluteus maximus lever arms based on the observed hip joint positions. Hamstrings and gluteus maximus PCA values were 42.4 mm² and 17.36 mm², respectively (Yamaguchi et al., 1990). H_d and GMd were determined for hip angular positions between 30° of flexion to 90° of extension (Duda, Brand, Freitag, Lierse, & Schneider, 1996; Nemeth & Ohlsen, 1985). Mean H_d and GMd values throughout the stance phase were 0.042 m and 0.047 m, respectively. The mean proportion of the hip extensor torque generated by the hamstrings (H_p) was equal to 0.63. The hamstrings force, H , was then calculated as:

$$H = H_p (\text{Het}) / H_d \quad (6)$$

where Het was the hip extensor torque. Hamstrings force was assumed to be zero when the hip torque was in the flexor direction, and hamstrings EMG supports this assumption (DeVita, 1994; Glitsch & Baumann, 1997). The direction of the hamstrings force was a line parallel to the femur at angle β to the tibia.

Quadriceps force was calculated from the observed net torque at the knee and the hamstrings and gastrocnemius forces. The observed net knee torque was a function of all muscles crossing the joint:

$$K_t = Q (Q_d) - H (H_d) - G (G_d) \quad (7)$$

where K_t was the net knee torque from inverse dynamics, Q , H , and G were the forces by the quadriceps, hamstrings, and gastrocnemius muscles, and Q_d , H_d , and G_d were the respective lever arms for the muscles at the knee based on the observed knee joint positions. Lever arm values were derived as described above from previous literature (Draganich, Andriacchi, & Andersson, 1987; Herzog & Read, 1993; Nisell & Ekholm, 1985; Smidt, 1973; Spoor & van Leeuwen, 1992; Spoor et al., 1990; Visser, Hoogkamer, Bobbert, & Huijing, 1990; Yamaguchi et al., 1990). The mean values for the three lever arms throughout the knee positions observed during the stance phase were, $Q_d = 0.035$ m, $H_d = 0.032$ m, and $G_d = 0.018$ m. The force in the quadriceps, Q , was then calculated as:

$$Q = [Kt + H (Hd) + G (Gd)] / Qd \quad (8)$$

The direction of Q , ϕ , was determined from the literature (Nisell, 1985; Yamaguchi & Zajac, 1989) and was also a function of the knee angle.

The three predicted forces along with the joint reaction forces were applied to the knee to determine the shear force, K_s :

$$K_s = G \sin \alpha - H \sin \beta + Q \sin \phi - K_z \sin \lambda + K_y \cos \lambda \quad (9)$$

where K_z and K_y were the vertical and horizontal knee joint reaction forces. K_s was positive when the shear force applied an anterior load to the tibia.

Peak muscle forces and muscle impulses and peak knee shear force and knee impulse were calculated from each trial and averaged across trials for each participant and condition. Impulses were calculated as the area under the muscle and knee joint force/time curves. The effects of the knee brace on peak forces and impulses were analyzed with a 2×2 , mixed factor (one repeated, one independent factor) analysis of variance (ANOVA). Two direct comparisons were performed: (a) ACL group, brace vs. no-brace condition; and (b) ACL group no brace vs. Control group no brace. Alpha level was set at 0.05.

Results

Hamstrings and quadriceps muscle forces during the stance phase of walking differed significantly between ACL and healthy persons (Figure 2; Table 1). Both groups had similar peak hamstrings force of about 1.29 BW ($p > .05$) at 5% into the stance phase. Compared to the healthy controls, however, those in the ACL group generated force

Table 1 Mean ($\pm$ SD) Muscle and Joint Forces and Impulses

	Healthy		ACL-No Brace		ACL-Brace
Hamstrings					
Peak force	1.27 (0.35)		1.31 (0.30)		1.29 (0.15)
Impulse	0.09 (0.06)	α	0.16 (0.08)	γ	0.23 (0.05)
Quadriceps					
Peak force	2.66 (0.71)	α	1.90 (0.23)	γ	1.54 (0.27)
Impulse	0.58 (0.15)	α	0.47 (0.19)	γ	0.41 (0.14)
Gastrocnemius					
Peak force	1.12 (0.17)		1.04 (0.06)		1.13 (0.12)
Impulse	0.28 (0.12)		0.26 (0.13)		0.31 (0.11)
Knee Shear Force					
Peak force	0.79 (0.21)	α	0.47 (0.15)	γ	0.34 (0.10)
Impulse	0.25 (0.04)	α	0.21 (0.09)	γ	0.17 (0.07)

Note: Force values in body weights; impulse values in Ns/kg body mass.

α = Healthy vs. ACL-No Brace, $p < .05$; γ = ACL-No Brace vs. ACL-Brace, $p < .05$.

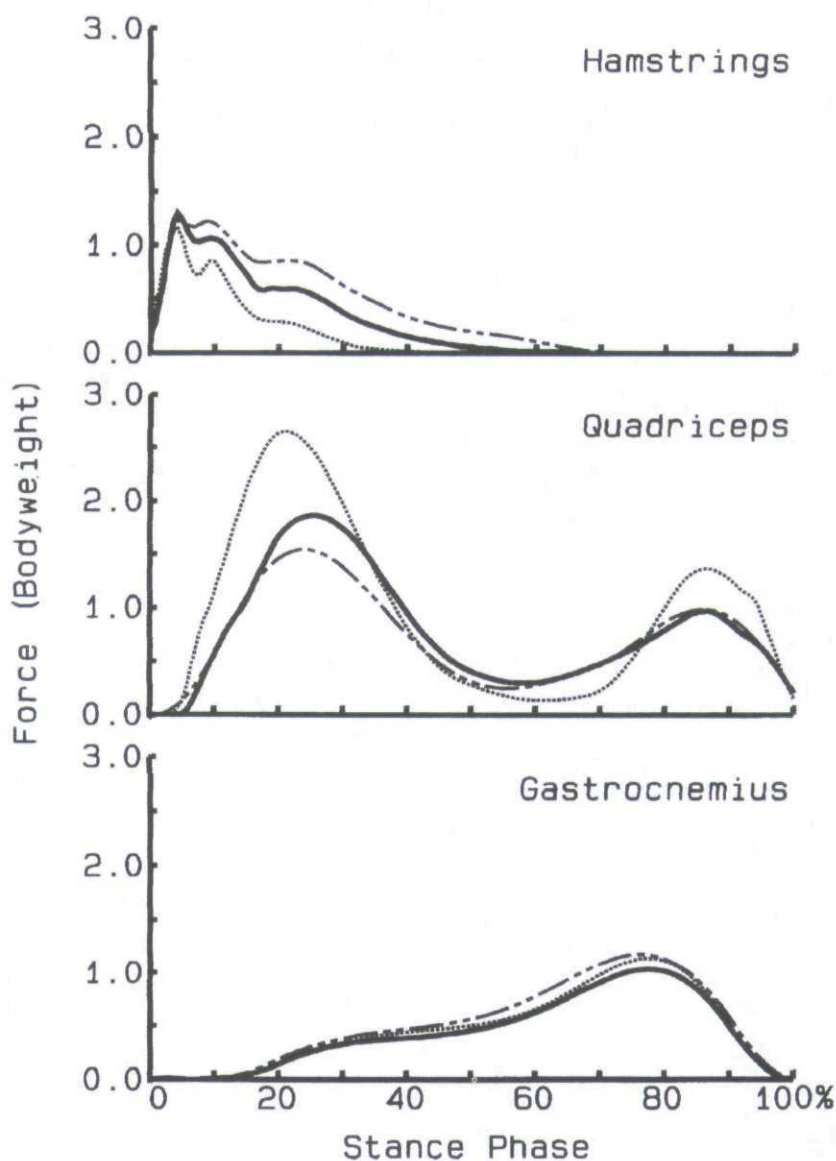


Figure 2 — Muscle forces during the stance phase of walking. Solid line = ACL—No brace; Dashed line = ACL—Brace; Dotted line = Healthy controls. Compared to controls, the ACL group walked with 78% more hamstrings impulse ($p < .02$) and 19% less quadriceps impulse ($p < .04$). Also, in the ACL group the functional knee brace produced an additional 43% increase in hamstrings impulse ($p < .04$) and an additional 13% decrease in quadriceps impulse ($p < .05$).

with their hamstrings longer into the stance phase and produced 78% more impulse ($p < .02$) with this muscle. Differences in quadriceps forces between groups were seen in the first half of the stance phase. Peak quadriceps force was 29% lower ($p < .04$) and quadriceps impulse was 19% lower ($p < .04$) in the ACL group compared to healthy controls. Gastrocnemius muscle forces did not differ significantly ($p > .05$) between groups or treatments.

The functional knee brace changed the lower extremity muscle forces during the stance phase in the ACL group. It produced an additional 43% increase in the impulse from the hamstrings ($p < .04$) in the ACL group compared to walking without the brace. The knee brace caused additional decreases in the quadriceps variables in the ACL group. Peak quadriceps force was reduced 19% ($p < .03$) and quadriceps impulse was reduced 13% ($p < .05$) in ACL participants walking with the brace vs. walking without it.

Differences in the muscle forces and the joint reaction forces (not reported) caused the anteroposterior shear force to be different between groups and between the brace and no-brace conditions for ACL participants (Figure 3; Table 1). As with the hamstrings and quadriceps forces, the changes in knee shear force occurred during the first half of the stance phase. Peak anterior knee shear force was 41% lower ($p < .01$) and shear impulse was 16% lower ($p < .05$) in ACL participants walking without the knee brace compared to healthy controls. The knee brace further reduced the peak knee shear force by 28% and the shear impulse by 19% in the ACL group (both $p < .04$).

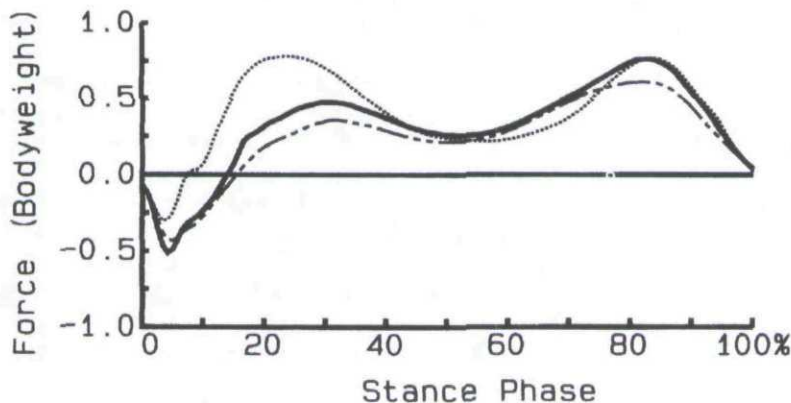


Figure 3 — Anteroposterior knee shear force during the stance phase of walking. Solid line = ACL-No brace; Dashed line = ACL-Brace; Dotted line = Healthy controls. Positive force was directed anteriorly onto the tibia. Peak force in early stance and impulse throughout stance were 41% lower ($p < .01$) and 16% lower ($p < .05$) in the ACL group vs. healthy controls. Also, in the ACL group the functional knee brace further reduced the peak knee shear force by 28% and impulse by 19% (both $p < .04$).

Table 2 Peak Muscle Force Predictions From Literature and Present Model

Source	Hamstrings	Quadriceps	Gastrocnemius
Anderson & Pandy (2001)	0.9	1.9	1.3
Collins et al. (1991)	1.1	1.1	1.0
Glitsch & Baumann (1997)	0.7	1.3	2.3
Hardt (1978)	1.5	3.7	1.0
Pedersen et al. (1987)	0.9	—	0.5
Seireg & Arvikar (1973)	1.2	1.7	—
Zatsiorsky et al. (1989)	1.1	3.9	1.2
Mean	1.1	2.3	1.2
<i>SD</i>	0.3	1.2	0.6
Present model	1.3	2.7	1.1

Note: All values in units of body weights and estimated from figures or taken from tables.

Discussion

Predictions of muscle and joint forces during human locomotion have the potential to identify fundamental biomechanical and neurological components to this activity. These predictions must be considered with caution, however, because the validation protocols for mathematical models in the literature have not been fully developed. We do not have direct measurements of the muscle and joint forces reported in the participants and thus cannot absolutely validate the results. However, we can provide a comparison between the current predictions from the healthy individuals and those from several other studies as an estimate of the validity of our procedures. We can also compare our triceps surae force with measured forces in the Achilles tendon.

The mean difference between previous and present predictions of the peak muscle forces during the stance phase of walking was only 18%, or 0.50 *SD*, and was deemed small (Table 2). Most important, the peak quadriceps force in the present participants differed by only 17%, or 0.35 *SD*, from the mean value reported in the literature. The shapes of the force curves were also similar to those reported in the literature (see references in Table 2). The hamstrings curve reached a peak value early in stance and declined to zero near midstance, the quadriceps curve was biphasic with peaks at 20% and 85% of stance, and the gastrocnemius curve began near zero at heel strike and reached a maximum value at 80% of the stance phase. Using a buckle-transducer, Komi et al. (Komi, Fukashiro, & Jarvinen, 1992) reported peak Achilles tendon forces of 3.6 BW in humans walking at 1.2 m/s, which agrees with our prediction of 3.5 BW for the triceps surae.

The restraining function of the ACL has been investigated with a number of mathematical models of the lower extremity, and many of these models were two-dimensional, sagittal plane analyses (Beynon, Yu, Huston, et al., 1996; Pandy & Shelburne, 1997). While the ACL does provide some restraint to transverse plane

rotations, its principle function rests in limiting anterior translation of the tibia on the femur, a sagittal plane function. Glitsch and Baumann (1997) recently showed that two- and three-dimensional models of the lower extremity produce similar knee joint forces during walking, particularly during the first half of the stance phase in which our principle results were observed. These favorable comparisons between our healthy participants and those in the literature and the planar function of the ACL led us to conclude that the present model of knee muscle and knee anteroposterior shear force was appropriate for the purpose of this study.

The ACL group produced more force with their hamstrings and less force with their quadriceps compared to the healthy group during the stance phase of gait. Experimental studies (Markolf, Gorek, Kabo, & Shapiro, 1990; Renstrom, Arms, Stanwyck, Johnson, & Pope, 1986) and theoretical studies (Beynon et al., 1996; Pandy & Shelburne, 1997) have shown that for knee joint positions used in the early stance phase of walking, i.e., 10–30° of flexion, hamstrings force decreased the load and strain applied to the ACL while quadriceps force increased the ACL load and strain. Indeed, the combined effect of the altered muscle forces in the ACL group compared to the healthy group was to reduce the anterior shear force at the knee by 41%. Therefore, it is clear that after ACL injury and bone–patellar tendon–bone reconstruction surgery, individuals adopt a healthier muscle coordination pattern that reduces load on the graft while still enabling them to perform the desired locomotor movement.

Previous research on individuals with ACL reconstruction reported larger net extensor torques at the hip and smaller net extensor torques at the knee during level walking (DeVita et al., 1998a), running (DeVita, Blakenship-Hunter, & Skelly, 1992), and ascending stairs (Kowalk et al., 1997) compared to healthy people. These studies concluded that ACL injury and surgery produce neuromuscular adaptations that protect the injured knee joint during locomotion, yet enable the individuals to perform the desired movements. The present results provide the first direct empirical support for this conclusion by demonstrating changes in muscle forces in individuals who have undergone ACL reconstruction.

The functional knee brace provided additional favorable muscular adaptations during the stance phase of walking. It further increased the total impulse generated by the hamstrings and further decreased the peak force and impulse generated by the quadriceps. We previously reported that the knee brace caused 7 of the present 9 participants to walk with ~5° less knee flexion, and at 3 weeks earlier in their rehabilitation compared to the present measurements (DeVita et al., 1998b). Similarly, the knee brace enabled the present participants to walk in a more erect position (~4°). This position reduced the external torque at the knee and therefore the force production required by the quadriceps during the first half of stance. This position would also lengthen the hamstrings compared to more flexed positions.

The increased muscle length may be one factor enhancing force production from the hamstrings during the first half of the stance phase. The 19% reduction in quadriceps impulse while wearing the functional knee brace was in close agreement with the ~20% reductions in quadriceps EMG with bracing in isokinetic and side-cutting tests in ACL-deficient persons (Branch et al., 1989; Wojtys et al., 1996). The increased impulse from the hamstrings while wearing the brace, however, contrasts with reduced hamstring EMG with bracing in these studies. We speculate that along with the hamstrings lengthening effect of the functional knee brace, it took additional

hamstrings force to compensate for the reduced quadriceps force in order to maintain a relatively normal walking pattern. A final interesting hypothesis can be drawn from these results. Because the functional knee brace changed the patterns of muscle force production, we hypothesize that with chronic use they may be useful in training specific motor patterns in persons with ACL reconstruction.

The critical effect of the functional knee brace was to reduce the peak anterior shear force and impulse on the knee joint by ~24%, with most of this reduction occurring during the first half of the stance phase. These results are analogous to those of Beynnon et al. (1997), who reported that a functional knee brace reduced ACL strain from an applied force while standing with flexed knees. Their results and ours combine to indicate that a functional knee brace can reduce ACL strain from an anterior shear force at the knee and can also reduce the anterior shear force during dynamic activity.

While the literature is replete with studies of the interactions of functional knee braces and ACL injury, there have been few studies identifying these interactions in individuals with ACL reconstruction. Most of the studies on functional knee braces have tested either healthy persons (e.g., Beynnon et al., 1997) or those with non-repaired ACL deficiency (e.g., Branch et al., 1989; Wojtys et al., 1996). We chose to apply the brace to individuals with recent ACL reconstruction because studies on animals suggest the graft is inherently weakest at about 6 to 12 weeks after surgery (Kohn & Rose, 1994; Rodeo, Arnoczky, Torzilli, Hidaka, & Warren, 1993). And studies on humans have shown that "rapid remodeling" (i.e., ligamentization) of the autograft does not begin until at least 8 weeks after the surgery (Rougaff, Shelbourne, Gerth, & Warner, 1993). We therefore propose that current research should focus on the application of functional knee braces to individuals with recent ACL reconstruction because the protective effects of the braces are most necessary at this time and *this population appears to be sensitive to the braces. The results of the present study lead us to conclude that a functional knee brace protects an ACL reconstructed with bone-patellar tendon-bone surgery during the stance phase of walking by altering muscle forces and reducing the anterior shear force applied to the knee joint in the early rehabilitation period.*

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